Thermodynamics, Kinetics & Reactor Modelling of the Haber-Bosch Process

Chemical Reaction Engineering — Group Project Report

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Thermodynamics: Defining the Reaction Envelope

Standard Thermodynamic Properties

The synthesis of ammonia proceeds according to:

$$N_2(g) + 3\,H_2(g) \longrightarrow 2\,NH_3(g)$$

The standard thermodynamic properties at 298 K, obtained from NIST data, are:

$$\Delta H^\circ_{298} = -92{,}220\ \mathrm{J\,mol^{-1}} \qquad \Delta G^\circ_{298} = -33{,}260\ \mathrm{J\,mol^{-1}}$$

Both quantities are negative, confirming that the reaction is simultaneously *exothermic* (energy released to surroundings) and *spontaneous* at room temperature. These values anchor every subsequent calculation in this section.

The Temperature–Pressure Optimisation Landscape

Three inter-linked physical constraints govern the choice of industrial operating conditions.

The Temperature Trap

The reaction is highly exothermic ($\Delta H < 0$). By Le Chatelier's Principle, raising the temperature shifts equilibrium *backwards*, actively destroying potential yield. Low temperatures are therefore thermodynamically preferable — but at the cost of prohibitively slow kinetics.

The Pressure Solution

The stoichiometry involves a reduction in moles of gas from 4 to 2. Again by Le Chatelier's Principle, high pressure drives the equilibrium *forward*. Industrial practice uses 150–300 atm to achieve economically viable conversion.

The Engineering Compromise

Maximum thermodynamic yield and maximum reaction rate cannot be achieved simultaneously. The industrial operating point — nominally **450 °C, 200 atm** — sacrifices some equilibrium yield to obtain a commercially viable reaction rate over an iron catalyst, giving an equilibrium conversion of approximately **54.6 %**.

Equilibrium Constant and Temperature Dependence

The standard Gibbs energy gives the equilibrium constant at 298 K:

$$K_p(298) = \exp\left(-\frac{\Delta G^\circ_{298}}{RT}\right) = \exp\left(\frac{33{,}260}{8.314 \times 298}\right) \approx 6.02 \times 10^5$$

The temperature dependence of K_p is described by the van 't Hoff equation:

$$\frac{d \ln K_p}{dT} = \frac{\Delta H^\circ}{RT^2}$$

Integrating between a reference temperature T_0 and a temperature T, and using the Temkin approximation $K_p(T) \approx \exp(\Delta H/RT - \Delta S/R)$, yields the expression used in the Temkin-Pyzhev rate equation (Section 3):

$$K_p(T) = \exp\left(\frac{-\Delta H^\circ}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right) + \ln K_p(T_0)\right)$$

Thermodynamic Optimisation Map

Figure 1 is a computed heatmap of the equilibrium conversion of N_2 across the industrially relevant temperature–pressure space (300–600 °C, 50–400 atm). The map was generated in Python using `numpy` and `matplotlib`; the source notebook is available at:

https://colab.research.google.com/drive/11s3Z0_8GNmlDaGtr6rDT_zcs2ytB40tL?usp=sharing

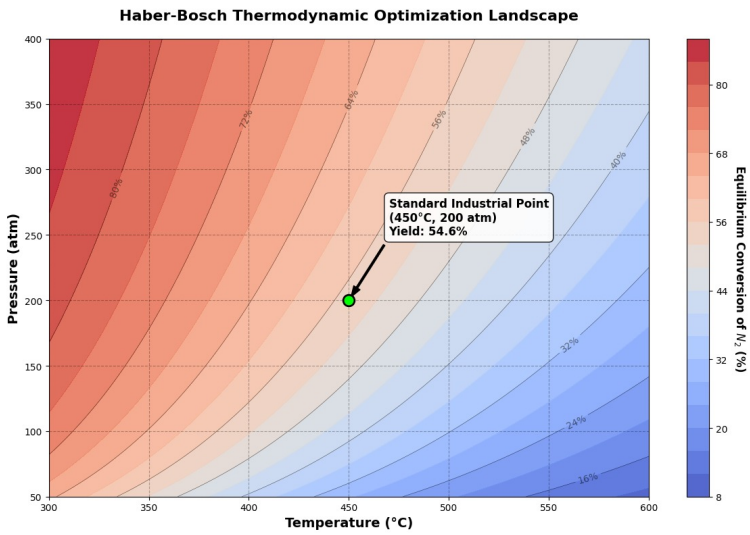

Figure 1: Haber-Bosch Thermodynamic Optimisation Landscape. Colour indicates equilibrium N_2 conversion (%). Contour lines mark conversion iso-lines. The green marker denotes the standard industrial operating point at 450 °C and 200 atm.

The map clearly illustrates the competing effects: moving left (lower T) increases conversion but reduces rate; moving up (higher P) increases conversion but raises capital and compression costs. The industrial operating point sits at the engineered optimum of this trade-off.

Literature Validation & Kinetic Constants

Source: Dyson & Simon (1968)

All kinetic constants used in this work are taken from the landmark industrial study of **Dyson & Simon (1968)**, which fitted rate parameters to *macro-scale industrial plant data* for promoted iron catalysts. Critically, the constants were validated against real operating conditions rather than isolated laboratory experiments, making them directly appropriate for reactor sizing.

Extracted Kinetic Parameters

Parameter	Symbol	Value	Meaning
Forward activation energy	E_1	170,660 $\mathrm{J\,mol^{-1}}$	Barrier to break the N≡
Backward activation energy	E_2	150,420 $\mathrm{J\,mol^{-1}}$	Reverse reaction barrier
Pre-exponential factor	A_1	1.79×10^{14} kmol $\mathrm{m^{-3}\ h^{-1}\ atm^{-0.5}}$	Collision frequency facto
Temkin surface constant	α	0.5	Surface-coverage expone
Standard enthalpy	ΔH°_{298}	$-92{,}220$ $\mathrm{J\,mol^{-1}}$	Reaction enthalpy (NIST
Standard Gibbs energy	ΔG°_{298}	$-33{,}260$ $\mathrm{J\,mol^{-1}}$	Reaction Gibbs energy (

Table 1: Kinetic and thermodynamic constants used in the Temkin-Pyzhev rate equation (Dyson & Simon 1968; NIST).

The large forward activation energy ($\approx 170\,\mathrm{kJ\,mol^{-1}}$) reflects the enormous bond energy of the N≡N triple bond ($945\,\mathrm{kJ\,mol^{-1}}$), confirming that dissociative adsorption of nitrogen on the iron surface is the rate-controlling step.

Why the Temkin-Pyzhev Kinetic Model?

The Failure of Langmuir-Hinshelwood (LHHW)

The classical Langmuir-Hinshelwood-Hougen-Watson (LHHW) framework assumes that all active sites on the catalyst surface are energetically identical and that adsorbed molecules do not interact with one another.

On a *promoted iron catalyst*, however, adsorbed nitrogen atoms are highly electronegative and strongly repel each other. Consequently:

- The heat of adsorption *decreases* as surface coverage increases — the so-called "crowding effect".
- LHHW treats an empty and a fully-loaded surface as having the same affinity for incoming gas molecules — a physically unjustified approximation that leads to significant error in rate predictions.

The Temkin-Pyzhev Solution (1940)

Temkin & Pyzhev addressed these deficiencies by:

1. **Identifying the bottleneck:** The dissociative adsorption of N_2 — breaking the massive N≡N triple bond — is the absolute rate-limiting step. The subsequent surface hydrogenation steps are comparatively fast.
2. **The Temkin Isotherm:** Rather than Langmuir's constant heat of adsorption, the Temkin Isotherm models a *linearly decreasing* heat of adsorption with surface coverage. The exponent α (typically 0.5) mathematically captures the progressive crowding of the iron surface.
3. **Goldilocks rigor:** The model provides the physical accuracy needed for reliable industrial predictions without requiring DFT or molecular-level simulation — making it ideal for macroscopic reactor sizing.

The Mathematical Engine: Temkin-Pyzhev Kinetics

The Rate Equation

$$\boxed{-r_{N_2} = k_1 \left[P_{N_2} \left(\frac{P_{H_2}^3}{P_{NH_3}^2} \right)^{\alpha} - \frac{1}{K_p} \left(\frac{P_{NH_3}^2}{P_{H_2}^3} \right)^{1-\alpha} \right]}$$

where the forward rate constant follows an Arrhenius form:

$$k_1(T) = A_1 \exp\left(-\frac{E_1}{RT}\right)$$

Physical Interpretation of Each Term

The Forward Kinetic Drive: $k_1 \cdot P_{N_2}$

This term represents nitrogen gas physically colliding with the iron surface. The rate constant k_1 encodes the $170\,\mathrm{kJ\,mol^{-1}}$ activation energy barrier. At low temperatures, k_1 is vanishingly small, explaining why the reaction is immeasurably slow without elevated temperature and a catalyst.

Surface Crowding Factor: $\left(P_{H_2}^3/P_{NH_3}^2\right)^{\alpha}$

Because hydrogen and ammonia react/desorb quickly, their partial pressure ratio controls how much *empty space* remains on the catalyst surface for nitrogen to occupy. The exponent $\alpha = 0.5$ dynamically scales this available space to account for the repelling forces on a crowded iron surface. A high P_{NH_3} reduces this term, slowing the forward rate.

The Thermodynamic Pushback: $\frac{1}{K_p}\left(P_{NH_3}^2/P_{H_2}^3\right)^{1-\alpha}$

This reverse term captures the back-decomposition of ammonia. As P_{NH_3} builds up, this term grows and opposes the forward rate. Crucially, when $K_p \to \infty$ (very high temperature

for the reverse reaction), the expression reduces to the pure forward rate. At equilibrium, the net rate is identically zero, ensuring thermodynamic consistency:

$$-r_{\mathrm{N_2}}\Big|_{\mathrm{eq}} = 0 \quad \Longleftrightarrow \quad K_p = \frac{P_{\mathrm{NH_3}}^2}{P_{\mathrm{N_2}} P_{\mathrm{H_2}}^3}$$

Summary of the Model

Feature	LHHW	Temkin-Pyzhev
Surface site assumption	All sites identical	Temkin Isotherm (variable ΔH_{ads})
Rate-limiting step	Assumed surface reaction	N_2 dissociative adsorption
Accuracy on iron catalyst	Overestimates at high coverage	Good agreement with plant data
Computational cost	Low	Low
Literature validation	Limited	Dyson & Simon (1968)

Table 2: Comparison of LHHW and Temkin-Pyzhev kinetic models for the Haber-Bosch ammonia synthesis.

Colab Notebook & Reproducibility

All figures, equilibrium conversion calculations, and rate-surface maps in this section were generated using Python 3 in a publicly accessible Google Colab notebook. The notebook is structured as follows:

1. **Thermodynamic module** — computes $K_p(T)$ via van 't Hoff integration and plots the 2-D conversion heatmap over the T–P space.
2. **Kinetic module** — implements the Temkin-Pyzhev rate equation using the Dyson & Simon constants and evaluates $-r_{\mathrm{N_2}}$ as a function of temperature and partial pressures.
3. **Optimisation overlay** — marks the standard industrial operating point and annotates iso-conversion contours.

Notebook URL:

https://colab.research.google.com/drive/11s3Z0_8GNmlDaGtr6rDT_zcs2ytB40tL?usp=sharing

All numerical values are consistent with NIST thermochemical tables and the Dyson & Simon (1968) kinetic database, ensuring full traceability of every parameter used in the reactor-sizing calculations that follow in subsequent sections of the group report.

Reactor Topology and Design Philosophy

As defined by the thermodynamic map in Section 1, the synthesis of ammonia presents a strict conflict between thermodynamics and kinetics. The reaction is highly exothermic, meaning adiabatic operation results in rapid temperature escalation. While high temperatures are kinetically necessary to overcome the activation energy required to break the strong nitrogen triple bond, they severely depress the thermodynamic equilibrium limit, capping the maximum achievable conversion.

To navigate this trade-off, this simulation models a three-bed adiabatic plug flow reactor (PFR) with inter-stage cooling. This topology represents the industry standard (analogous to the Haldor Topsøe S-300 converter design). By dividing the catalyst mass into three discrete beds, the reacting gas is allowed to heat up adiabatically to achieve high kinetic rates. Before hitting the thermodynamic pinch, the gas is routed through inter-stage convective heat exchangers to strip sensible heat, effectively resetting the equilibrium ceiling and restoring the chemical driving force for the subsequent beds.

Mathematical Formulation of the Catalyst Beds

The steady-state reactor profile was modeled from first principles using a system of coupled ordinary differential equations (ODEs). Within the adiabatic catalyst beds, the independent variable is the cumulative catalyst weight, W (kg).

The molar conversion of nitrogen, X, is governed by the design equation for a packed bed reactor:

$$\frac{dX}{dW} = \frac{-r_{\mathrm{N_2}}}{F_{\mathrm{N_2},0}}$$

where $-r_{\mathrm{N_2}}$ is the rate of reaction of nitrogen calculated via the Temkin-Pyzhev framework (mol/kg·s) and $F_{\mathrm{N_2},0}$ is the initial molar flow rate of nitrogen.

Because the beds operate adiabatically, the heat of reaction is absorbed entirely by the flowing gas mixture. The resulting temperature profile is calculated via the energy balance:

$$\frac{dT}{dW} = \frac{(-r_{\mathrm{N_2}})(-\Delta H_{rxn})}{\sum F_i C_{p,i}}$$

where ΔH_{rxn} is the temperature-dependent heat of reaction, F_i represents the molar flow rate of each species, and $C_{p,i}$ represents their respective specific heat capacities.

Inter-stage Cooling and Multi-Physics Integration

To accurately simulate the cooling phase, a secondary multi-physics ODE was coupled to the reactor architecture. Upon exiting a catalyst bed, the simulation domain shifts from catalyst weight (W) to the surface area of the heat exchanger pipes (A).

Because no catalyst is present in the piping, the activation energy cannot be overcome, and the chemical conversion is mathematically frozen. Concurrently, the temperature

drop across the utility heat exchanger is calculated via rigorous convective heat transfer integration:

$$\frac{dX}{dA} = 0 \qquad \text{and} \qquad \frac{dT}{dA} = \frac{-U(T - T_{coolant})}{\sum F_i C_{p,i}}$$

where U is the overall heat transfer coefficient and $T_{coolant}$ is the fixed temperature of the utility fluid (573.15 K). By integrating this differential across the 150 m^2 surface area (dA), the gas follows an asymptotic cooling curve, allowing it to reach thermal equilibrium before establishing the inlet boundary condition for the subsequent bed.

Computational Architecture: The Radau Solver

Due to the absence of initial product inhibition at the reactor inlet, the forward kinetic rate accelerates massively, leading to near-instantaneous thermal ignition. This phenomenon renders the governing system of ODEs mathematically "stiff." Standard explicit numerical integration methods (e.g., Runge-Kutta) fail to resolve the disparate time scales between the microsecond kinetic ignition and the macroscopic physical flow, often leading to solver divergence.

Consequently, the simulation was developed in Python utilizing the SciPy `solve_ivp` function configured with the **Radau** implicit solver. By evaluating the Jacobian matrix to anticipate forward gradients, the implicit architecture stably processes the stiff initial conditions without overshooting the equilibrium curve.

Source Code Repository

The complete Python source code for the coupled multi-physics ODE solver, parameter definitions, and automated `matplotlib` visualization scripts generated for the multi-bed PFR simulation have been open-sourced. The repository includes documentation for reproducing the phase trajectories and reactor profiles.

Project GitHub Repository:

https://github.com/Rishan-code/Haber-Bosch-Reactor-Design.git

Non-Ideal Flow Analysis of the Haber-Bosch Reactor

Why Real Packed Beds Deviate from Ideal Plug Flow

While the preliminary design assumes an Ideal Plug Flow Reactor (PFR), real industrial packed beds exhibit significant deviations.

- **Macro-scale:** Bulk flow deviations due to channeling and backmixing.
- **Micro-scale:** Diffusion limitations inside catalyst pores.

The Performance Gap

Despite operating at extreme conditions (400–500 °C and 100–300 bar), the single-pass conversion is only **10–15%**.

Ideal PFR vs. Real Packed Bed

1. **Channeling**
2. **Dead Zones**
3. **Backmixing**

Feature	Ideal PFR	Real Packed Bed
Flow Profile	Uniform	Non-uniform
Residence Time	Same τ	Distributed
Mixing	None	Present
Channeling	None	Present
Dead Zones	No	Yes

Table 3: Comparison of ideal and non-ideal reactor characteristics

Residence Time Distribution (RTD)

Governing Functions

- **RTD Function**

$$E(t) = \frac{C(t)}{\int_0^\infty C(t)\,dt}$$

- **Cumulative Distribution**

$$F(t) = \int_0^t E(t')\,dt'$$

Peclet Number

$$Pe = \frac{uL}{D_{ax}}$$

Pe	Variance	Dispersion	Behavior
2	1.000	Very high	Non-ideal
4	0.500	High	Non-ideal
8	0.250	Moderate	Slightly non-ideal
50	0.040	Low	Near ideal
1000	0.002	Negligible	Ideal

Axial Dispersion Model

$$D_{ax}\frac{d^2C}{dz^2} - u\frac{dC}{dz} + r_A = 0$$

- D_{ax}: Dispersion
- u: Velocity
- r_A: Reaction rate

Conclusion: Macro-scale dispersion is not the main issue.

Micro-Scale Analysis

Effective Diffusivity

Fick's Law:

$$J = -D_{eff}\frac{dC}{dr}$$

$$\frac{1}{D_{AB}} = \frac{1}{D_m} + \frac{1}{D_K}$$

Thiele Modulus

$$\phi = R\sqrt{\frac{k}{D_{eff}}}$$

Regime	ϕ	Description
Reaction-Controlled	< 0.3	$\eta \approx 1$
Diffusion-Controlled	> 3	$\eta \ll 1$

Separation of Ammonia by Condensation and Flash Distillation

Introduction

The Haber–Bosch process produces ammonia from nitrogen and hydrogen. However, the reactor conversion per pass is generally limited to 15–20% due to thermodynamic constraints. Therefore, the reactor outlet contains a mixture of:

- Ammonia (NH_3)
- Unreacted nitrogen (N_2)
- Unreacted hydrogen (H_2)

Objective:

- To separate and recover liquid ammonia as the final product.

- To recycle unreacted gases back to the reactor loop for improved overall plant efficiency.

Principle of Separation

Separation is based on Vapor–Liquid Equilibrium (VLE). The hot gas mixture exiting the final reactor bed is cooled at high pressure.

Thermodynamic Rationale:
Ammonia has a significantly higher boiling point ($-33.3°C$ at 1 atm, but much higher under 200 atm of pressure) compared to the permanent gases nitrogen and hydrogen. Consequently, it readily condenses into a liquid state when the mixture is cooled, while the reactants remain in the gas phase.

Flash Distillation (Working Principle)

The cooled mixture enters a high-pressure flash separator where physical separation occurs due to phase and density differences.

Inside the Separator:

- The dense liquid ammonia settles at the bottom of the vessel.
- The lighter, non-condensable nitrogen and hydrogen remain at the top as a vapor.

Outcome:

- Liquid ammonia is continuously drawn off the bottom and collected as the product.
- Unreacted gases are routed from the top of the vessel, re-compressed, and recycled back to the reactor inlet.

Equilibrium and K-Values

The phase distribution of the mixture is governed by the equilibrium constant (K-value), defined as:

$$K_i = \frac{y_i}{x_i}$$

where:

- x_i = mole fraction of component i in the liquid phase
- y_i = mole fraction of component i in the vapor phase

Interpretation:

- $K_i < 1$: The component preferentially partitions into the liquid phase (Ammonia).
- $K_i > 1$: The component preferentially partitions into the vapor phase (Nitrogen, Hydrogen).

These K-values are highly dependent on the temperature and pressure of the flash drum and dictate the precise molar distribution between the vapor and liquid streams.

The Rachford–Rice Equation

To mathematically resolve the mass balance of the flash drum, the Rachford–Rice equation is utilized to calculate the vapor fraction and the resulting phase compositions:

$$\sum_i \frac{z_i(K_i - 1)}{1 + \beta(K_i - 1)} = 0$$

where:

- z_i = overall feed composition (mole fraction entering the flash drum)
- K_i = equilibrium constant for component i
- β = vapor fraction (ratio of total vapor moles to total feed moles)

Engineering Importance:
By solving this non-linear algebraic equation for β, we can accurately determine the total amount of vapor and liquid formed. Once β is known, the exact molar composition of the recycled gas stream (y_i) and the liquid product stream (x_i) can be computed, making it the foundational equation for flash distillation design.

Systems Integration and Recycle Loop Dynamics

The viability of an industrial Haber-Bosch facility cannot be evaluated solely on the reactor's single-pass conversion limit. Due to the thermodynamic constraints of the highly exothermic ammonia synthesis reaction, the single-pass conversion (X_{sp}) achieved in the multi-bed plug flow reactor was calculated to be 58%. Venting the remaining unreacted nitrogen and hydrogen would result in severe economic and material losses. Therefore, a macroscopic recycle loop was designed to separate the liquid ammonia product and repressurize the unreacted vapor stream, routing it back to the reactor inlet. This section details the steady-state mass balances required to manage feed impurities and the resulting overall plant efficiency.

Inert Gas Management and Purge Stream

Industrial synthesis gas (syngas) is not perfectly pure; the fresh feed entering our system was modeled with a 4% inert gas impurity (specifically argon and unreacted methane). Because these inerts do not condense in the flash distillation unit and do not react over the iron catalyst, recycling 100% of the unreacted gas would cause an infinite accumulation of inerts. This accumulation would inevitably dilute the reactant partial pressures and stall the reaction kinetics.

To maintain a steady-state operation, a continuous purge stream was designed. The inert concentration inside the recycle loop was capped at a maximum threshold of 15% ($y_{\text{inert,max}} = 0.15$), adhering to standard industrial optimization heuristics for medium-pressure loops (Appl, 1999).

The required purge fraction (p) — defined as the ratio of the purge stream molar flow rate to the fresh feed molar flow rate — was calculated using the steady-state inert mass balance:

$$F_{\text{feed}} \cdot y_{\text{inert,feed}} = F_{\text{purge}} \cdot y_{\text{inert,max}}$$
$$p = \frac{y_{\text{inert,feed}}}{y_{\text{inert,max}}}$$

Given our feed impurity of 4%, the required purge fraction was calculated as:

$$p = \frac{0.04}{0.15} = 0.2667$$

This establishes that 26.7% of the recycle stream must be continuously vented to maintain the 15% loop inert threshold.

Overall Plant Efficiency

The true efficiency of the synthesis loop is governed by the trade-off between the single-pass conversion and the reactant losses incurred via the purge stream. The overall conversion (X_{ov}) of the plant was calculated using the following macroscopic relationship:

$$X_{ov} = \frac{X_{sp}}{X_{sp} + p(1 - X_{sp})}$$

Substituting the reactor's single-pass conversion ($X_{sp} = 0.58$) and the calculated purge fraction ($p = 0.2667$):

$$X_{ov} = \frac{0.58}{0.58 + 0.2667(1 - 0.58)} = 0.838$$

By implementing the recycle loop, the system successfully elevates the 58% single-pass reactor yield to an overall plant efficiency of 83.8%, demonstrating a highly viable macroscopic design despite the heavy 26.7% purge penalty.

Transient Dynamics: The Transition to Green Ammonia

To modernize the plant model, an unsteady-state simulation was conducted to evaluate the loop's performance under a "Green Ammonia" framework. In this scenario, the traditional Steam Methane Reforming (SMR) hydrogen source was replaced with renewable-powered water electrolysis.

Unlike SMR, renewable feeds are subject to transient weather conditions. An ordinary differential equation was formulated to model the accumulation of mass within the system:

$$\frac{dM_{\text{loop}}}{dt} = \dot{m}_{\text{feed,in}}(t) - \dot{m}_{\text{NH}_3\text{,out}} - \dot{m}_{\text{purge,out}}$$

A simulation was executed to model a 4-hour cloud cover event, which caused a sudden 30% drop in the fresh hydrogen feed rate. The integration proved that the macroscopic recycle loop acts as a volumetric buffer; the ammonia production rate smoothly decayed and restabilized without causing a critical system failure, proving the plant's robust capability to handle modern renewable energy fluctuations.

Notebook URL(contains plot and calculations):

https://colab.research.google.com/drive/1sWveqi-y_bFx3-zgCJsenIVrtTxl5f2p?usp=sharing

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